

MARCOMATTIA MOCELLIN

Senior Software Engineer | Technical Lead | Software Architect | Industry 4.0 Specialist

+39 349 136 5331 | marcomattia.mocellin@gmail.com | Castelfranco Veneto, Italy

PROFESSIONAL OVERVIEW

Senior Software Engineer and Technical Lead with more than 10 years of experience in industrial software, embedded systems, electronics, metrology, and Industry 4.0. Proven experience coordinating cross-functional technical activities and delivering software and hardware solutions from concept and prototyping through industrialization, deployment, and customer support. Strong background in C#, .NET, WPF, embedded Linux, industrial communication protocols, software architecture, and project coordination. Currently serving in technical coordination, project management, and AI implementation leadership roles within the voestalpine Group.

CORE EXPERTISE

Technical Leadership | Software Architecture | Industry 4.0 | Industrial Automation | Embedded Systems | Artificial Intelligence Integration | Project Management | Digital Manufacturing | Industrial Internet of Things (IIoT) | Industrial Connectivity | Cross-Functional Coordination | C# | .NET | Embedded Linux | Yocto Linux | ARM Platforms | REST APIs | MES Integration | CAN Bus | MQTT

LEADERSHIP HIGHLIGHTS

Technical Coordinator | Coordinate software, hardware, firmware, and integration activities for a voestalpine Group Industry 4.0 platform since May 2022.

Project Manager | Coordinate the development of a new industrial hardware interface solution since June 2022.

AI Tech Lead | Provide technical leadership for Group AI implementation projects since May 2025.

Cross-Functional Delivery | Define protocols, validate technical specifications, supervise software quality and field testing, and support customers and project staging.

WORK EXPERIENCE

SOFTWARE ENGINEER | TECHNICAL LEAD | voestalpine Böhler Welding Arc Technology S.r.l., Onara, Italy NOV 2020 - PRESENT

- Develop software and connected solutions supporting welding equipment maintenance, firmware operations, production integration, data acquisition, configuration, and Industry 4.0 use cases.
- Serve as Technical Coordinator since May 2022 for a voestalpine Group Industry 4.0 platform, aligning software, hardware, firmware, and integration activities.
- Serve as Project Manager since June 2022 for the development of a new hardware interface solution.
- Serve as Tech Lead since May 2025 for Group AI implementation projects.
- Modernized the Weld@Net software suite by converting applications to C# and applying SOLID principles where feasible; maintained and optimized legacy C++ components, achieving performance improvements of up to 1200% in selected areas.
- Act as Technical Lead for weldNet Process Manager; developed middleware and software for an NXP i.MX 6ULL-based interface, including .NET applications on ARM and Yocto Linux integration.
- Define communication protocols, supervise code quality and field testing, review technical specifications, produce documentation, support demonstrations, and assist customers.

TEST ENGINEER | SOFTWARE ENGINEER | FPT Industrie S.p.A., Santa Maria di Sala, Italy OCT 2016 - NOV 2020

- Developed software and integration solutions for measuring systems, CNC machine tools, and Industry 4.0 manufacturing environments.
- Implemented applications connecting production management and planning software (MES) with numerical controls, including Heidenhain DNC and LSV-2, Selca, and Balluff libraries.
- Contributed to machine qualification, testing, Industry 4.0 compliance, and retrofit connectivity using IIoT devices.
- Built C# WPF and DevExpress applications, REST services, and SQL/SQLite integrations; worked with Fanuc, Siemens, Heidenhain, ServiceStack, and Modbus TCP/IP.

ELECTRONICS ENGINEER | CNR-IFN Padova / EXTRACAL Project, Padua, Italy SEP 2012 - SEP 2016

- Led technical activities for an optical metrology system designed to compensate geometric errors in large boring and milling machines operating in industrial environments.
- Owned project activities from optical and mathematical design through prototype development, laboratory testing, field validation, industrialization, and post-installation support.
- Designed optical paths and mathematical models for machine axes exceeding 15 metres, with accuracy requirements in the tens of micrometres.
- Coordinated mechanical, electrical, automation, and metrology specialists; maintained reporting and partner communication.
- Supported patent drafting and analysis activities that contributed to five registered patents worldwide and co-authored two scientific publications.

SELECTED PROJECTS

Weld@Net Software Suite | Modernization of applications supporting firmware updates, troubleshooting, production data collection, and weld validation.

weldNet Process Manager | Industrial welding-process platform based on an NXP i.MX 6ULL interface, .NET on ARM, Yocto Linux, and distributed communication protocols.

MARES 4.0+ | Industry 4.0 platform for manufacturing data acquisition and production management; developed CNC drivers, alarm management, and machine keep-alive features.

EXTRACAL | Optical calibration system for medium and large machine tools; designed, tested, and deployed the optical subsystem from prototype through customer support.

EDUCATION

Master Degree in Electronic Engineering | University of Padua, Padua, Italy 2008 - 2012
Quality and Reliability curriculum with additional studies in optoelectronics. Thesis: Shape from Silhouette Reconstruction: Self-Calibration for X-Ray Applications.

FrontierLab@OsakaU Certificate | Osaka University, Osaka, Japan 2011
International research exchange focused on terahertz imaging. Research topic: New Terahertz Radiation Detector Design.

Bachelor Degree in Electronic Engineering | University of Padua, Padua, Italy 2003 - 2008
Thesis: Safety Application PLCs.

SKILLS

Software Development | C#, .NET Framework, .NET, WPF, ASP.NET Core, Razor Pages, REST APIs, C++, MFC, Object-Oriented Programming, SOLID principles

Embedded and Industrial Systems | Embedded Linux, Yocto Linux, ARM, NXP i.MX 6ULL, CAN Bus, MQTT, UDP/TCP, Modbus TCP/IP, IIoT, CNC and MES integration

Tools and Platforms | GitLab Enterprise, NuGet, WiX Toolset, PowerShell, Bash, SQLite, ServiceStack, DevExpress, MATLAB, National Instruments libraries, MikroTik RouterOS

Leadership and Delivery | Technical leadership, software architecture, project management, technical coordination, stakeholder management, requirements management, release planning, technical governance, customer-facing delivery, specifications, documentation, workflow supervision

PUBLICATIONS

Optical System for the Calibration and Verification of Correct Axis Positioning in Medium-Big Sized Milling Boring Machines | Proceedings of SPIE 9525, 2015. DOI: 10.1117/12.2184620.

Optical Device for the Improvement of Positioning Accuracy in Large Machine Tools | Proceedings of SPIE 8788, 2013. DOI: 10.1117/12.2020288.

LANGUAGES

Italian | Native English | C2 reading, listening, and writing; C1 spoken production and interaction Japanese | A1